Part 3 | Base salary

Question 1. One-sided test and Type I error

Let μ be the population mean base salary. Test $\mathbf{H_0}$ with μ ≤ $130,000 against $\mathbf{H_a}$ with μ > $130,000. Reject H_0 if the mean of 50 observations exceeds $144,000.

Using the full dataset's sample SD as the planning estimate, the standard error is $60,185.29/√50 = **$8,511.49**. At the null boundary μ = $130,000, the cutoff is (144,000 − 130,000)/8,511.49 = 1.64484 standard errors above the mean.

α = P(Z > 1.64484) = 0.050002 ≈ 5.00%.

Question 2. Null sampling distribution

The curve is centered at $130,000 and spans μ_0 ± 8 SE ($61,908.12 to $198,091.88). The shaded right tail gives the probability of rejecting at the null boundary.

Figure 1 shows a probability-density histogram of 50,000 sample means simulated from the normal null model in MATLAB. The smooth curve is the theoretical null density. The red line marks $144,000, and the shaded area under the curve to its right is α.

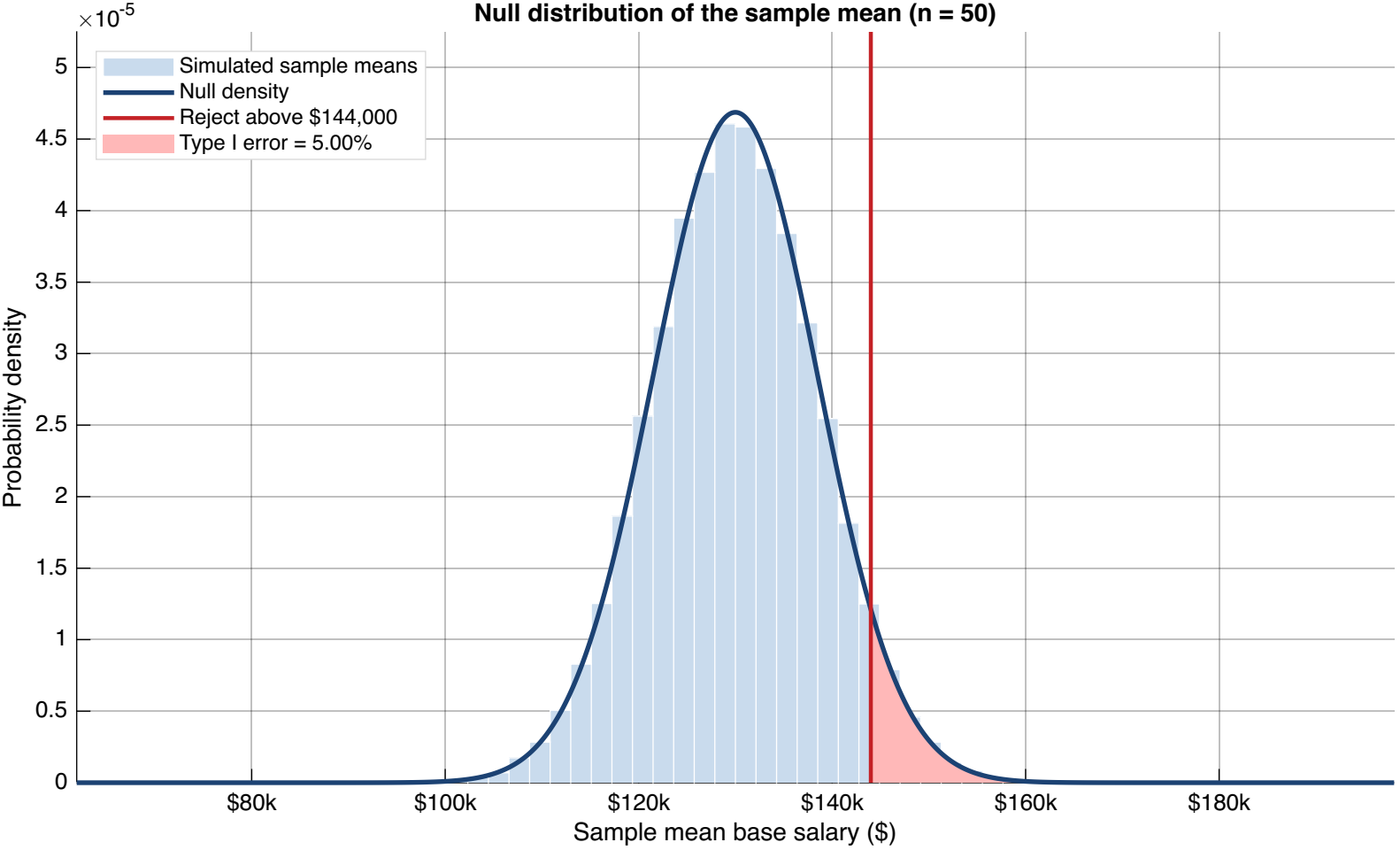
Run `part3/part3.m` in MATLAB to regenerate Figure 1.

Question 3. Power at a true mean of $150,000

The standardized effect size is **d** = (150,000 − 130,000)/60,185.29 = **0.3323**. Under the normal approximation, the probability that the sample mean exceeds $144,000 at this alternative is **75.96%**.

The one-sample, right-tailed t power at n = 50 and α = 0.050002 is **74.93%**. MATLAB computes this power using the noncentral t distribution. The t procedure uses a standard deviation estimated from the 50 observations, which accounts for the difference in power.

Data: Engineering_Salaries_2021_10000.csv, base_salary column. ChatGPT assisted with MATLAB syntax, numerical verification, and preparation of the figure and write-up.


Null distribution of the sample mean (n = 50)
×10^-5
Simulated sample means
Null density
Reject above $144,000
Type I error = 5.00%
Probability density
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
$80k
$100k
$120k
$140k
$160k
$180k
Sample mean base salary ($)